CS450

Structure of Higher Level Languages

Lecture 07: foldr, looping last-to-first

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Exercises on lists

Searching

Element in list?

Spec

```
(require rackunit)
(check-true (member? "d" (list "a" "b" "c" "d")))
(check-false (member? "f" (list "a" "b" "c" "d")))
```

Solution

```
(define (member? x l)
  (match l
    [(list) #f]
    [(cons h l)
      (define result (member? x l))
      (or (equal? x h) result)]))
```

Solution in Python

```
def member(x, l):
  for h in l:
    if h == x:
      return True
  return False
```

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Element in list?

Spec

```
(require rackunit)
(check-true (member? "d" (list "a" "b" "c" "d")))
(check-false (member? "f" (list "a" "b" "c" "d")))
```

Alternative solution

```
(define (member? x l)
  (match l
    [(list) #f]
    [(cons h _) #:when (equal? x h) #t]
    [(cons _ l) (member? x l)]))
```

Solution in Python

```
def member(x, l):
  for h in l:
    if h == x:
      return True
  return False
```

Prefix in list?

Spec

```
(require rackunit)
(check-true (string-prefix? "Racket" "R")) ; available in standard library
(check-true (match-prefix? "R" (list "foo" "Racket")))
(check-false (match-prefix? "R" (list "foo" "bar")))
```

Solution

```
(define (match-prefix? p l)
  (match l
    [(list) #f]
    [(cons h _) #:when (string-prefix? h p) #t]
    [(cons _ l) (match-prefix? p l)]))
```

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Can we generalize the search algorithm?

```
; Example 1
(define (member? x l)
  (match l
    [(list) #f]
    [(cons h _) #:when (equal? h x) #t]
    [(cons _ l) (member? x l)]))
```

```
; Example 2
(define (match-prefix? p l)
  (match l
    [(list) #f]
    [(cons h _) #:when (string-prefix? h p) #t]
    [(cons _ l) (match-prefix? p l)]))
```

Solution

```
(define (exists? found? l)
  (match l
    [(list) #f]
    [(cons h _) #:when (found? h) #t]
    [(cons _ l) (exists? found? l)]))
```

```
; Example 1
(define (member? x l)
  (exists?
    (lambda (y) (equal? x y)) l))
; Example 2
(define (match-prefix? x l)
  (exists?
    (lambda (y) (string-prefix? y x))) l)
```

Removing elements from list

Remove zeros from a list

Spec

```
(require rackunit)
(check-equal? (list 1 3 4) (remove-0s (list 0 1 3 0 4)))
(check-equal? (list 1 2 3) (remove-0s (list 1 2 3)))
```

Solution

```
(define (remove-0s l)
  (match l
    [(list) (list)]
    [(cons h l) #:when (not (equal? h 0))
      (cons h (remove-0s l))]
    [(cons _ l) (remove-0s l)]))
```

Solution in Python

```
def remove_0s(l):
  result = []
  for h in l:
    if h != 0:
      result.append(h)
  return result
```

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Can we generalize this functional pattern?

Original

```
(define (remove-0s l)
  (match l
    [(list) (list)]
    [(cons h l) #:when (not (equal? h 0))
      (cons h (remove-0s l))]
    [(cons _ l) (remove-0s l)]))
```

Generalized

```
(define (filter keep? l)
  (match l
    [(list) (list)]
    [(cons h l) #:when (keep? h)
      (cons h (filter keep? l))]
    [(cons _ l) (filter keep? l)]))

;; Usage example
(define (remove-0s l)
  (filter
    (lambda (x) (not (equal? x 0))) l))
```

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Concatenate two lists

Concatenate two lists

Implement function `(append l1 l2)` that appends two lists together.

Spec

```
(check-equal?
  (append (list 1 2) (list 3 4))
  (list 1 2 3 4))
```

Solution

```
(define (append l1 l2) ; l2 = (list 4 5 6)
  (match l1            ; l1 = (list 1 2 3)
    [(list) l2]
    [(cons h l1)                      ; h = 1   l1 = (list 2 3)
      (define result (append l1 l2)) ; result = (append '(2 3) '(4 5 6))
      ; result = (list 2 3 4 5 6)
      (cons h result)]))
      ; output = (list 1 2 3 4 5 6)
```

Generalizing order-preserving loops

An order-preserving recursion pattern

1. Case `(list)` (handle-base)
2. Case `(cons h l)` (handle-step)
3. Recursive call handles "smaller"

```
(define (rec l)
  (match l
    [(list)    handle-base]
    [(cons h l)
      (define result (rec l))
      (handle-step h result)]))
```

Example 1

```
(define (map f l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define result (map f l))
      (cons (f h) result)]))
      ; = (handle-step h result)
```

Example 2

```
(define (append l1 l2)
  (match l1
    [(list) l2]
    [(cons h l1)
      (define result (append l1 l2))
      (cons h result)]))
```

A note about side-effects

- We need to be mindful when implementing `map` as the order of side-effects may matter.
- The **standard** implementation of `map` invokes `f` from left-to-right.
- Our implementation of `map` invokes `f` from right-to-left.
- The reason we implement r-t-l is to allow for generalization with `foldr`
- In terms of code, to obtain an l-t-r ordering, call `f` before recursing
- Run `(map displayln (list 1 2 3))` on either version to observe the difference

Side-effects from right-to-left

```
(define (map f l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define result (map f l))
      (cons (f h) result)]))
```

Side effects from left-to-right

```
(define (map f l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define x (f h))
      (define result (map f l))
      (cons x result)]))
```

An order-preserving recursion pattern

Searching

```
(define (exists? found? l)
  (match l
    [(list) #f]
    [(cons h _) #:when (found? h) #t]
    [(cons _ l) (exists? found? l)]))
```

Following the recursion pattern

```
(define (exists? found? l)
  (match l
    [(list) #f]
    [(cons h l)
      (define result (exists? found? l))
      (or (found? h) result)]))
```

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An order-preserving recursion pattern

Removing

```
(define (filter keep? l)
  (match l
    [(list) (list)]
    [(cons h l) #:when (keep? h)
      (cons h (filter keep? l))]
    [(cons _ l) (filter keep? l)]))
```

Following the recursion pattern

```
(define (filter keep? l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define result (filter keep? l))
      (if (keep? h)
        (cons h result) result)]))
```

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Implementing this recursion pattern

Implementing this recursion pattern

Recursive pattern for lists

```
(define (rec l)
  (match l
    [(list) base-case]
    [(cons h l)
      (define result (rec l))
      (handle-step h result)]))
```

Fold right reduction

```
(define (foldr handle-step base-case l)
  (match l
    [(list) base-case]
    [(cons h l)
      (define result (foldr handle-step base-case l))
      (handle-step h result)]))
```

```
# In Python
def foldr(handle_step, base_case, l)
  result = base_case
  for h in reversed(l):
    result = handle_step(h, result)
  return result
```

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Implementing map with foldr

```
(define (map f l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define result (map f l))
      (cons (f h) result)]))
```

Solution

```
(define (map f l)
  (foldr
    ; step: how do you build the next result
    (lambda (h result) (cons (f h) result))
    ; what to return when the list is empty
    (list)
    ; iterate/match over l
    l))
```

```
# Python pseudo-code
result = []
for h in reversed(l):
  # result = cons(f(h), result)
  result.append(f(h))
```

Implementing append with foldr

```
(define (append l1 l2)
  (match l1
    [(list) l2]
    [(cons h l1)
      (define result (append l1 l2))
      (cons h result)]))
```

Solution

```
(define (append l1 l2)
  (foldr
    ; step: add the element to the list being built
    (lambda (h result) (cons h result))
    ; base-case: start with list l2
    l2
    ; iterate/match over l1
    l1))
```

Implementing filter with foldr

```
(define (filter keep? l)
  (match l
    [(list) (list)]
    [(cons h l)
      (define result (filter keep? l))
      (if (keep? h) (cons h result) result)]))
```

Solution

```
(define (filter keep? l)
  (foldr
    ; handle-step
    (lambda (h result) (if (keep? h) (cons h result) result))
    ; base-case
    (list)
    ; iterate/match over l
    l))
```

Implementing exists? with foldr

```
(define (exists? found? l)
  (match l
    [(list) #f]
    [(cons h l)
      (define result (exists? found? l))
      (or (found? h) result)]))
```

Solution

```
(define (exists? found? l)
  (foldr
    ; handle-step
    (lambda (h result) (or (found? h) result))
    ; base-case
    #f
    ; iterate over l
    l))
```